ANTI-D IMMUNOGLOBULIN

RhD-negative mothers

Within 72 h of sensitising event

CHEMICAL NAME Human anti-D (Rh_o) immunoglobulin — IgG antibodies against RhD antigen from pooled plasma of immunised RhD-negative donors

CLASS Specific human immunoglobulin (prevents Rh alloimmunisation)

TRADE NAMES RhoGAM · Rhophylac · D-Gam · WinRho

THERAPEUTIC USES / INDICATIONS

- Prevention of RhD sensitisation in RhD-negative women: routine antenatal prophylaxis (28 weeks); after delivery of an RhD-positive baby.
- After sensitising events: miscarriage (>12 weeks or surgical), ectopic pregnancy, termination, abdominal trauma, amniocentesis, CVS, external cephalic version, antepartum haemorrhage.
- Immune thrombocytopenia (RhD-positive non-splenectomised, IV — specialist).

ADMINISTRATION

- **IM** into deltoid (preferred) — or IV (some products, e.g. Rhophylac).
- Give as soon as possible and within 72 h of sensitising event (some benefit up to 10 days).
- Blood product: consent, record batch number.
- Kleihauer test after delivery/large events to quantify fetomaternal haemorrhage (FMH).

DOSAGES

- **Routine antenatal (RAADP):** 1,500 IU at 28 weeks (single dose) or 500 IU at 28 and 34 weeks.
- **Postnatal (baby RhD-positive):** at least 500 IU within 72 h (more if Kleihauer shows FMH >4 mL; 125 IU per mL of fetal red cells).
- **Sensitising events <20 weeks:** 250 IU; ≥20 weeks: 500 IU + Kleihauer.

PHARMACOKINETICS

- **Peak:** IM 2–3 days. **t½:** ~3–4 weeks (can be detected in maternal blood for weeks — note when interpreting antibody screens).

NURSING MANAGEMENT

- Confirm mother's RhD-negative status and antibody screen; cord blood group after birth.
- Give within 72 h; document batch, dose, time, site; informed consent.
- Kleihauer test after delivery and events ≥20 weeks.
- Identify all potentially sensitising events in RhD-negative women.

PHARMACODYNAMICS

- Binds and clears **RhD-positive fetal red cells** from maternal circulation before the mother's immune system recognises them → prevents formation of anti-D antibodies → protects future pregnancies from haemolytic disease of the fetus and newborn (HDFN).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site pain, tenderness.
- Low-grade fever, headache.
- Mild chills.

ADVERSE EFFECTS (SERIOUS)

- Hypersensitivity/anaphylaxis (rare).
- Intravascular haemolysis (IV use in ITP).
- Theoretical infection transmission.

CONTRAINDICATIONS & CAUTIONS

- RhD-positive women (prophylactic use).
- RhD-negative women already sensitised (anti-D antibodies present; not effective).
- Hypersensitivity to human immunoglobulin; IgA deficiency with antibodies.

KEY INTERACTIONS

- **Live vaccines (MMR, varicella) postpartum:** may reduce response; still give, but check rubella immunity later (or give at same time/separate sites per guidance).

PATIENT EDUCATION

- Explains it protects future babies from blood problems.
- Report any bleeding, abdominal trauma or pain in pregnancy promptly — you may need another dose.
- Keep a record of doses.

KEY POINTS

- Prevents RhD sensitisation in RhD-negative women.
- After any sensitising event.
- 1,500 IU at 28 weeks; ≥500 IU within 72 h postpartum.
- Kleihauer test guides extra dosing.